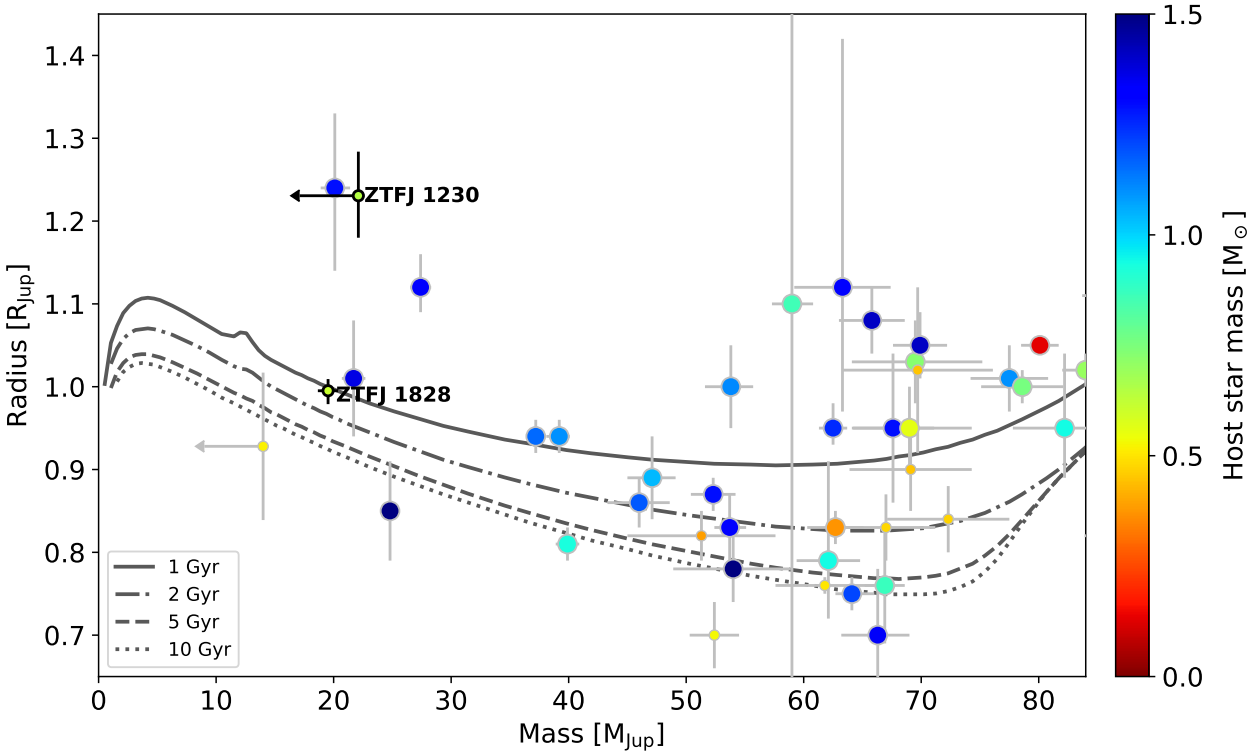

Figure 10. Measured masses and radii of transiting substellar objects with ages >1 Gyr (Littlefair et al. 2014; Parsons et al. 2017b; Vanderburg et al. 2020; Casewell et al. 2020a; van Roestel et al. 2021; Grieves et al. 2021; Psaridi et al. 2022; Vowell et al. 2023, 2025; Lin et al. 2023; El-Badry et al. 2023; Page et al. 2024; Henderson et al. 2024; Schaffenroth et al. 2014, 2015, 2021), where the mass of the host star is indicated by the colour of the point. White dwarf and sdB hosts are shown as small points, while main sequence star hosts are shown as large points. ZTF J1230−2655 and ZTF J1828+2308 are highlighted as bold points. Also shown are a number of different evolutionary tracks from Marley et al. (2021) for different ages. The masses of the substellar objects in ZTF J1230−2655 and WD 1856+534 are upper limits, as indicated by the arrows.

old age of the system. The brown dwarfs in the non-eclipsing systems GD 1400 (Casewell et al. 2024) and NLTT 5306 (Amaro et al. 2023) may also be over-inflated. However, the three other eclipsing white dwarf plus brown dwarf systems with precise mass and radius measurements show good agreement with theoretical models (Littlefair et al. 2014; Parsons et al. 2017b; van Roestel et al. 2021). The situation is equally mixed for the population of brown dwarfs around main sequence stars, around half of which appear to be over-inflated. In order to clearly highlight this large variation we used the catalogue of transiting brown dwarfs from Grieves et al. (2021) and removed any systems from the that were younger than 1 Gyr or where there was no good estimate for the age. This means that all the systems plotted in Figure 10 should be at or below the 1 Gyr track in order to have measured radii consistent with theoretical models, but many clearly lie significantly above this track. There is no clear trend with over-inflation and host star mass (hence host star temperature) either, although given that the orbital periods of these systems range from around a day up to more than 100 days the actual irradiation levels vary significantly from object to object.

In order to see if the most over-inflated objects are also the most irradiated, we calculated the time-averaged equilibrium temperature of all of the objects plotted in Figure 10 using the relation in Méndez & Rivera-Valentín (2017), assuming zero albedo. Note that this neglects the intrinsic luminosity of the brown dwarf, and assumes that it is in thermal equilibrium with the external irradiation. These values are highlighted in Figure 11, where the highly irradiated objects are clear. However, there is no clear correlation between the irradiation levels and over-inflation, with some highly irradiated objects both oversized (e.g. KELT-1b, Siverd et al. 2012) and consistent with models (e.g. SDSS J1205-0242B, Parsons et al. 2017b). Likewise there are many objects with low levels of irradiation that are both oversized (e.g. NGTS-19b, Acton et al. 2021) and consistent with models (e.g. LHS 6343C, Johnson et al. 2011). However, the equilibrium temperature only gives a limited view of how these objects are irradiated, since the way the brown dwarf atmosphere responds to the radiation will depend on the wavelength of the radiation. Hotter stars (in particular white dwarfs) emit more of their light at ultraviolet wavelengths, which tends to be absorbed in the higher layers of the brown dwarf's atmosphere potentially leading to temperature inversions (Lothringer & Casewell 2020), while cooler stars emit significant amounts of infrared light, which can pass deeper into the atmosphere before being absorbed, potentially heating (or preventing the cooling of) their interiors (Komacek & Youdin 2017).

Interestingly, the over-inflation of the brown dwarf in ZTF J1230−2655 is consistent with the trend that, in white dwarf plus brown dwarf binaries, it is the oldest and least irradiated objects that appear to be the most over-inflated (French et al. 2024). The over-inflation may therefore be the result of the cumulative effect of

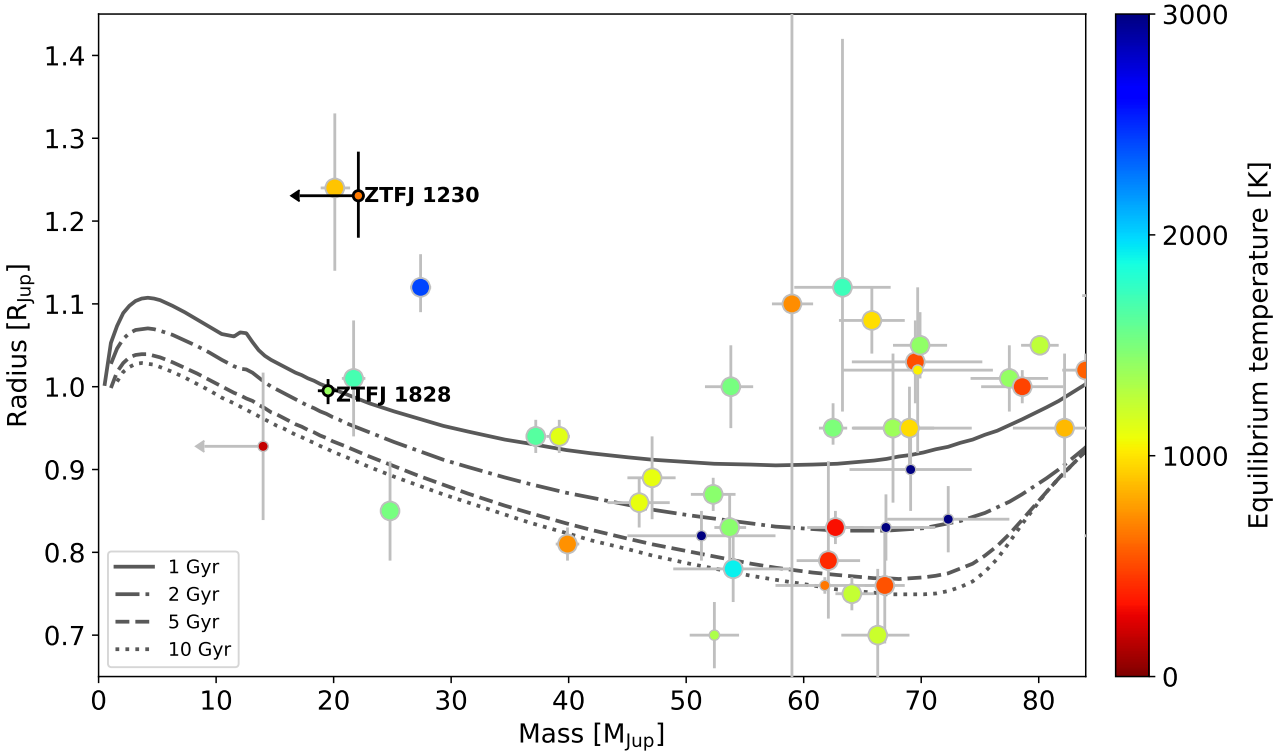

Figure 11. Same as Figure 10, except points are coloured by the equilibrium temperature of the brown dwarf. Note that this temperature ignores the intrinsic luminosity of the brown dwarf, it is purely the equilibrium temperature due to irradiation.

irradiation from the white dwarf over a very long time scale slowing down the contraction of these brown dwarfs and the younger systems simply haven't been irradiated for long enough for this effect to be detectable (Casewell et al. 2020b). However, the brown dwarf in ZTF J1230−2655 is so over-inflated that simply delaying the normal contraction is likely not enough to explain its current radius, which is more consistent with a 0.5 Gyr old brown dwarf (despite a total age of 1.5−2 Gyr, see the next section). Additional mass-radius measurements for close white dwarf brown dwarf binaries should help test this hypothesis, if sufficient numbers (over a wide range of white dwarf cooling ages) can be found and measured.

Alternatively, the brown dwarf in ZTF J1230−2655 may be over-inflated due to possessing a thick cloud layer or high metallicity (Burrows et al. 2011). Unfortunately, since we did not detect any flux from the brown dwarf itself we cannot place any useful limits on the conditions in the atmosphere of this object. Given their low masses, direct detection of the brown dwarfs in either of the systems in this paper will likely be challenging, but might be possible with the James Webb Space Telescope.

4.2 Evolution

Given their extremely small separations and circular orbits, both ZTF J1230−2655 and ZTF J1828+2308 are likely the result of CE evolution. There is some debate if the transiting planet candidate around WD 1856+534 is also a post-CE system or if it migrated into its present orbit after the formation of the white dwarf. Given that WD 1856+534 is part of a triple system this migration channel seems plausible (Muñoz & Petrovich 2020; O'Connor et al. 2021), but the CE channel may still be possible (Lagos et al. 2021). Indeed, the existence of two additional systems with similarly low mass companions in close orbits around white dwarfs raises the possibility that there does indeed exist a channel for creating such systems via CE evolution. We note that neither of our two new systems appear to form part of a higher-order system at present and there are no common proper motion companions to either system in Gaia DR3 and therefore dynamical scattering or migration seems unlikely in these cases. Indeed, to migrate the brown dwarf in ZTF J1828+2308 from several AU (wide enough to avoid the giant star phases) down to its present day separation of just 0.8 $R_\odot$ in only 160 Myr (the cooling age of the white dwarf) seems extremely challenging. We therefore discuss the evolution of these two systems assuming that they have passed through a CE phase, although as we will see, there are still challenges with this interpretation.

It is also worth noting that the low Roche-lobe filling factors of both systems mean that it is unlikely these are period bounce CVs that have temporarily detached. The short cooling age of the white dwarf in ZTF J1828+2308 also supports this, since it would take far longer than the 160 Myr cooling age for a standard CV to evolve to this phase. While it may be possible to detach period bounce CVs to these kinds of fill factors via the late appearance of a strong white dwarf magnetic field (Schreiber et al. 2023), neither of the white dwarfs in these systems have a detectable magnetic field (the lack of any obvious Zeeman splitting in the hydrogen Balmer lines rules out any field stronger than a MG or so), so this mechanism could not have occurred in these cases. The white dwarf mass in both systems is also

Table 2. Stellar and binary parameters for ZTF J1230−2655 and ZTF J1828+2308. Distances are from Bailer-Jones et al. (2021). For ZTF J1828+2308 $K_{\rm BD}$ is the corrected value of the centre of mass velocity of the brown dwarf, based on the emission line velocity $K_{\rm emis} = 343.4 \pm 1.2\,{\rm km\,s^{-1}}$. The white dwarf in ZTF J1230−2655 is forced to follow the mass-radius relationship of Bédard et al. (2020). The quoted RLFFs are the volume-average Roche lobe fill factors.

Value	Unit	ZTF J1230−2655	ZTF J1828+2308
RA	hours	12:30:16.59	18:28:48.77
Dec	deg	−26:55:51.34	+23:08:38.39
Gaia G	mag	18.851 ± 0.003	17.996 ± 0.003
Gaia $G_{\rm BP} - G_{\rm RB}$	mag	0.20 ± 0.04	−0.11 ± 0.02
D	pc	193 ± 9	204 ± 5
$P_{\rm orb}$	days	0.235 977 66(95)	0.112 006 70(87)
T_0	BMJD(TDB)	60432.047 880 5(32)	60431.149 404 4(2)
$K_{\rm WD}$	${\rm km\,s^{-1}}$	<9	10.5 ± 1.1
$\gamma_{\rm WD}$	${\rm km\,s^{-1}}$	11.7 ± 1.5	20.5 ± 0.9
$K_{\rm BD}$	${\rm km\,s^{-1}}$	-	367.0 ± 1.2
$\gamma_{\rm BD}$	${\rm km\,s^{-1}}$	-	−12.1 ± 0.9
i	deg	86.7 ± 0.3	88.9 ± 0.4
a	$R_\odot$	1.400 ± 0.012	0.838 ± 0.002
$M_{\rm WD}$	$M_\odot$	0.646 ± 0.017	0.610 ± 0.004
$R_{\rm WD}$	$R_\odot$	0.0123 ± 0.0002	0.0131 ± 0.0002
WD $\log(g)$	-	8.07 ± 0.05	8.05 ± 0.05
$T_{\rm eff,WD}$	K	10000 ± 110	15900 ± 75
$\tau_{\rm cool,WD}$	Gyr	0.726 ± 0.050	0.162 ± 0.003
$M_{\rm BD}$	$M_\odot$	<0.0211	0.0186 ± 0.0008
$R_{\rm BD}$	$R_\odot$	0.126 ± 0.005	0.102 ± 0.002
RLFF	-	0.59	0.77

somewhat lower than is typically found in CVs (e.g. McAllister et al. 2019; Pala et al. 2022). Therefore we consider both systems to be pre-CVs and the current stellar masses are the same as at the end of the CE phase.

While the exact details of the CE phase are still poorly understood, it is expected that drag forces acting on the brown dwarf once it has entered the atmosphere of the giant star will cause energy to be transferred from the orbit of the binary into the envelope. If the envelope gains a sufficient amount of energy it will be ejected, leaving behind a much tighter binary. However, this transfer of energy from the orbit to the envelope may not necessarily be very efficient and so only some fraction of the lost orbital energy is actually used to unbind the envelope, typically parametrized as $\alpha_{\rm CE}$. A low value of $\alpha_{\rm CE}$ implies that only a small fraction of the lost orbital energy is actually used to unbind the envelope, therefore the binary must lose more orbital energy overall in order to eject the envelope. This results in very close binaries emerging from the CE phase or even mergers if the envelope fails to be ejected. To date, studies of both white dwarf plus M dwarf binaries and white dwarf plus brown dwarf binaries have shown that almost all present day systems can be reconstructed with a low value of $\alpha_{\rm CE} \simeq 0.2 - 0.4$ (Zorotovic et al. 2010; Toonen & Nelemans 2013; Camacho et al. 2014; Cojocaru et al. 2017; Zorotovic & Schreiber 2022). Given the low masses (hence low orbital energies) of the brown dwarfs in our systems, they offer a particularly powerful test of whether this low value of $\alpha_{\rm CE}$ is universal for these kinds of binaries.

We reconstructed the evolution of our systems following the method of Zorotovic & Schreiber (2022), who investigated the CE phase for number of white dwarf plus brown dwarf binaries. In brief, we used the algorithm of Zorotovic et al. (2010), which first uses the measured stellar and binary parameters to correct for any loss of angular momentum (hence reduction in orbital period) since the end of the CE phase. Due to the very low masses of the brown dwarfs it is assumed that only gravitational radiation has reduced the orbital period since the end of the CE phase. The algorithm then searches over a large grid of single star evolutionary tracks to identify any giant stars where the core mass is equal to the measured white dwarf mass. The orbital separation at the onset of the CE phase is then determined using the total mass and radius of the giant star, the mass of the brown dwarf (which we assume does not change) and Roche geometry. The algorithm then uses the energy formalism for CE from Webbink (1984), which relates the energy required to eject the envelope (i.e. its binding energy, $E_{\rm bind}$) and the orbital energy released by the binary ($\Delta E_{\rm orb}$),

$$E_{\rm bind} = \alpha_{\rm CE} \Delta E_{\rm orb}, \tag{4}$$

$$E_{\rm bind} = -\frac{G M_1 M_{1,e}}{\lambda R_1}, \tag{5}$$

$$\Delta E_{\rm orb} = \frac{1}{2} G M_{1,c} M_2 \left(\frac{1}{a_i} - \frac{1}{a_f} \right), \tag{6}$$

where M_1, $M_{1,c}$ and $M_{1,e}$ are the total, core and envelope mass of the giant star (i.e. $M_{1,c}$ is the mass of the present day white dwarf), R_1 is the radius of the giant star, M_2 is the mass of the brown dwarf, a_i ($= R_1$) and a_f are the initial and final orbital separations. The structural parameter λ is related to the mass distribution of the giant star's envelope and was calculated as in Claeys et al. (2014). This calculation also allows including additional energy sources that reduce the binding energy of the envelope and hence make it easier to expel, e.g. hydrogen recombination energy. However, we did not include these in our reconstructions in order to test whether these low mass systems can still be reconstructed without the need for extra energy sources beyond the orbital energy of the binary, which appears to be the case for other white dwarf plus brown dwarf binaries (Zorotovic & Schreiber 2022). $\alpha_{\rm CE}$ is treated as a free parameter and we initially fixed the metallicity to the solar value ($Z = 0.02$).

The results of this reconstruction are shown in Figure 12. For each model we show the total age of the system (the sum of the pre-CE lifetime and the current cooling age), the initial mass of the white dwarf progenitor and the mass of the progenitor at the onset of the CE phase. For ZTF J1230−2655 we assumed a mass of the brown dwarf of 0.0211 $M_\odot$, our measured upper mass limit. In Figure 12 we distinguish models where the CE took place on the AGB before the first thermal pulse (blue) and after the first thermal pulse (i.e. on the thermally pulsing AGB - TP-AGB, grey). Given the high white dwarf masses we found no solutions where the white dwarf progenitor was on the first giant branch (FGB). For ZTF J1230−2655 we only find solutions where the CE occurred during the TP-AGB. Moreover, all of these solutions also require the white dwarf progenitor to have lost a significant fraction of its total mass before the CE phase started. The situation is similar for ZTF J1828+2308, although AGB progenitors are possible for very high values of $\alpha_{\rm CE}$.

To be consistent with previous constraints of $\alpha_{\rm CE} \simeq 0.2 - 0.4$ would require that both of our systems underwent a CE phase when the white dwarf progenitor was on the TP-AGB and had already lost around a third to half of its original mass. This is perhaps not too surprising a result, given the low masses of the brown dwarfs, since the binding energy of the envelope is already quite low at this phase, allowing even a low mass companion to survive. However, there are potential issues with this interpretation. As pointed out by Belloni